

Esophagus and Stomach

Department of Anatomy

The abdomen (4)

First Part

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Lecture Points

- **Overview of the Digestive System.**
- **Gastrointestinal Functions.**
- **Introduction of Esophagus (Location and course).**
- **Relation of esophagus.**
- **abdominal esophagus.**
- **esophageal constrictions.**
- **Blood supply of the esophagus.**
- **Innervation of esophagus.**
- **Lymphatic drainage of esophagus.**
- **Function of esophagus and Gastroesophageal Sphincter.**

- **Stomach**
- **Function of the Stomach.**
- **Location and Description.**
- **Covering with peritoneum**
- **Covering with peritoneum and lesser omentum.**
- **Parts of the Stomach.**
- **Cardia Opening.**
- **Pylorus.**
- **Muscular layers.**
- **The stomach bed.**
- **Nerve supply of the stomach.**
- **Arterial supply stomach.**
- **Venous drainage of the stomach.**
- **Lymph Drainage of the stomach**

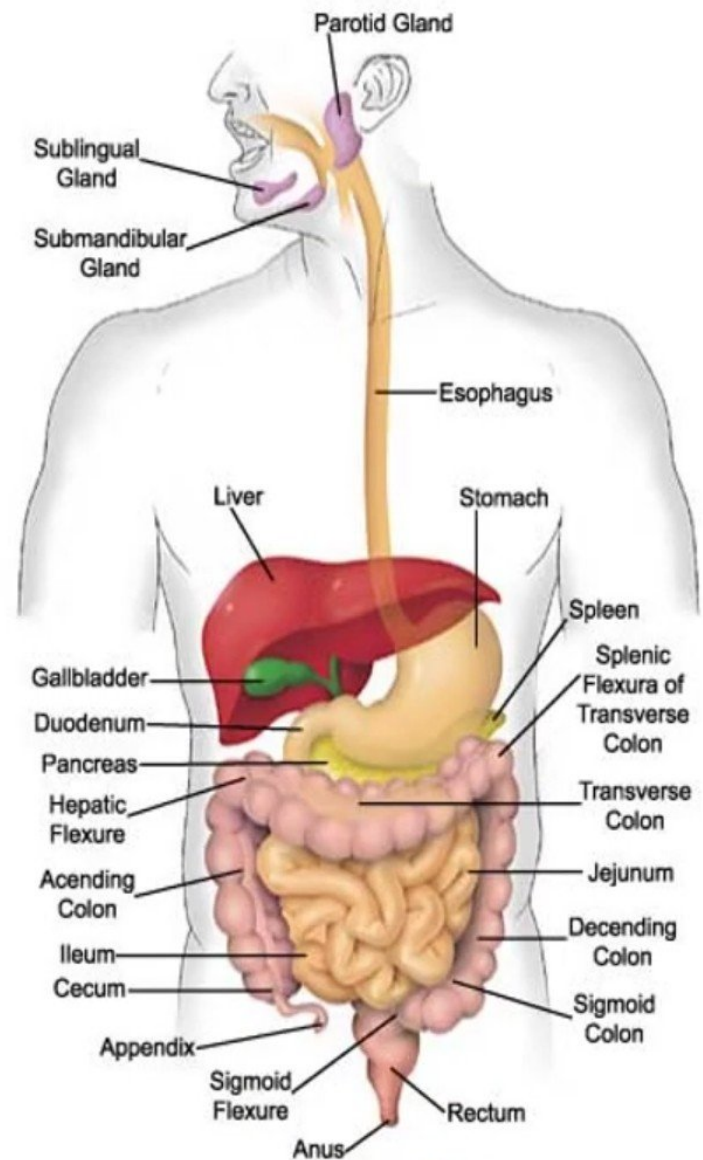
Overview of the Digestive System

Digestive Tract:

Mouth, pharynx, and esophagus, stomach, small intestine, and large intestine (colon).

Accessory Organs:

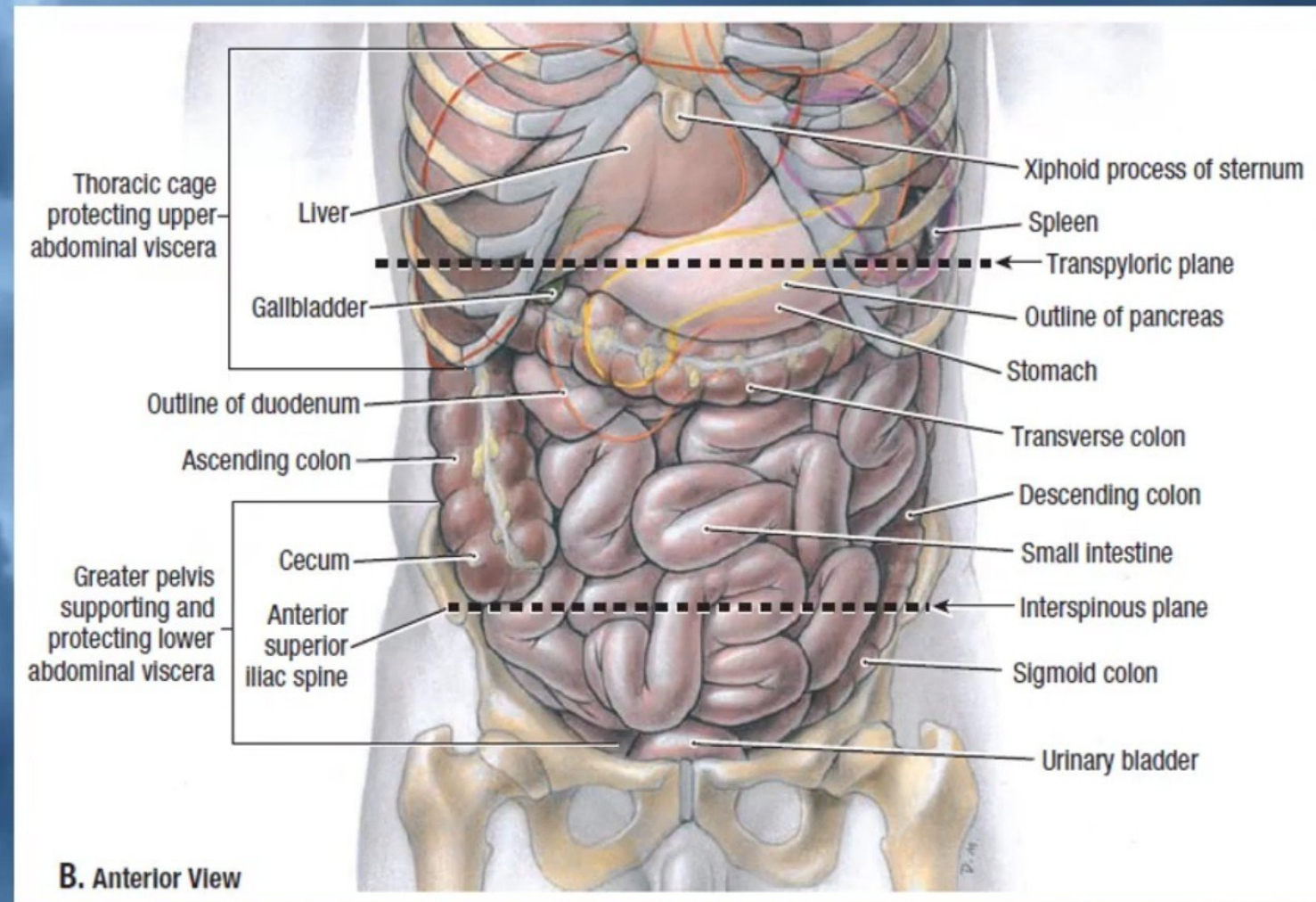
Teeth, tongue, salivary glands, pancreas, liver, and gallbladder



Gastrointestinal Functions

1) Digestion:
secrete enzymes which break down the complex food materials into simple form.

2) Absorption:
convey the end products to the bloodstream.



When learning about any organ in gastrointestinal system must be Know?

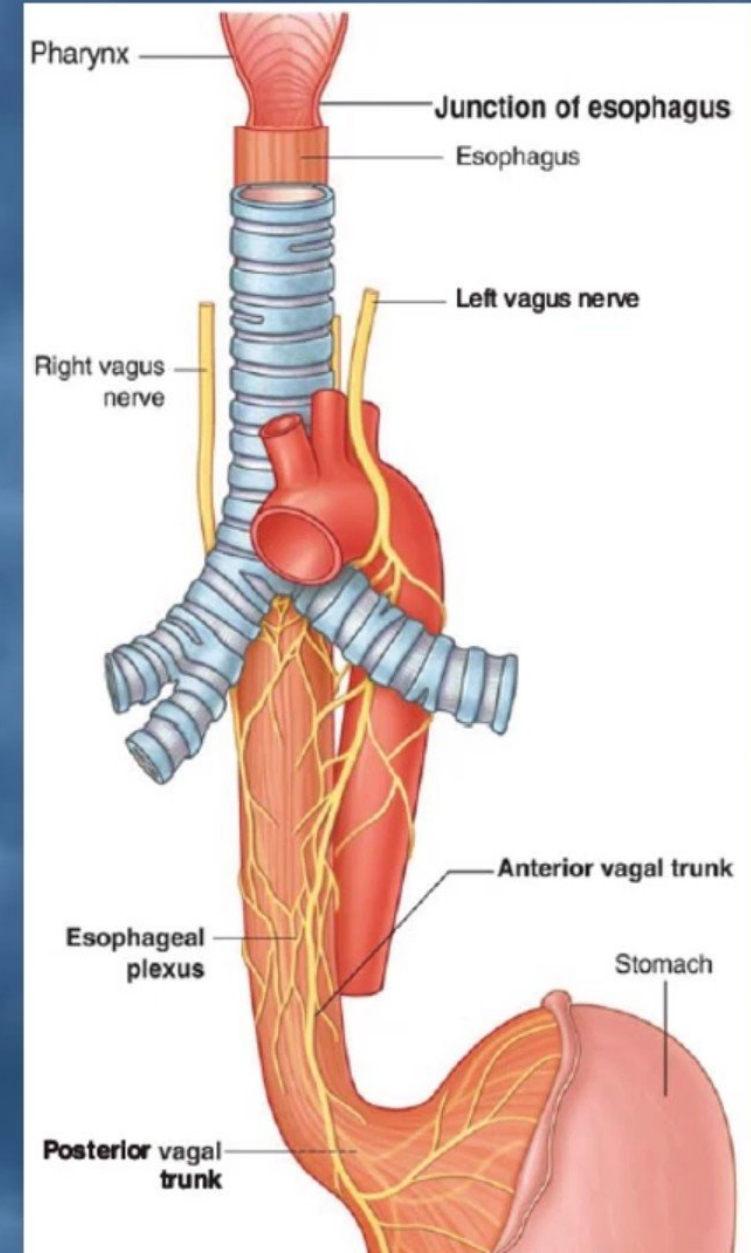
- ✓ Introduction of organ (Location and course).
- ✓ Covering with peritoneum and related ligament.
- ✓ Parts of organ.
- ✓ Relation of organ .
- ✓ Blood supply of the organ (Arterial and venous).
- ✓ Innervation of organ (sympathetic and parasympatetic).
- ✓ Lymphatic drainage of organ.
- ✓ Function of organ.
- ✓ Clinical correlation.

Introduction of Esophagus and course

- It is a **muscular tube** (approximately 10 in.) (25 cm long).
- It extends from the **pharynx** to the **stomach**.
- It is long that continuous above with **the laryngeal part of the pharynx opposite the sixth cervical vertebra**.
- It **passes through the diaphragm** at the level of **the 10th thoracic vertebra** to join the stomach.

In the neck:

- The esophagus lies **in front of** the **vertebral column**.
- **laterally**, it is related to the **lobes of the thyroid gland**.
- **anteriorly**, it is in contact with the **trachea** and **the recurrent laryngeal nerves**.



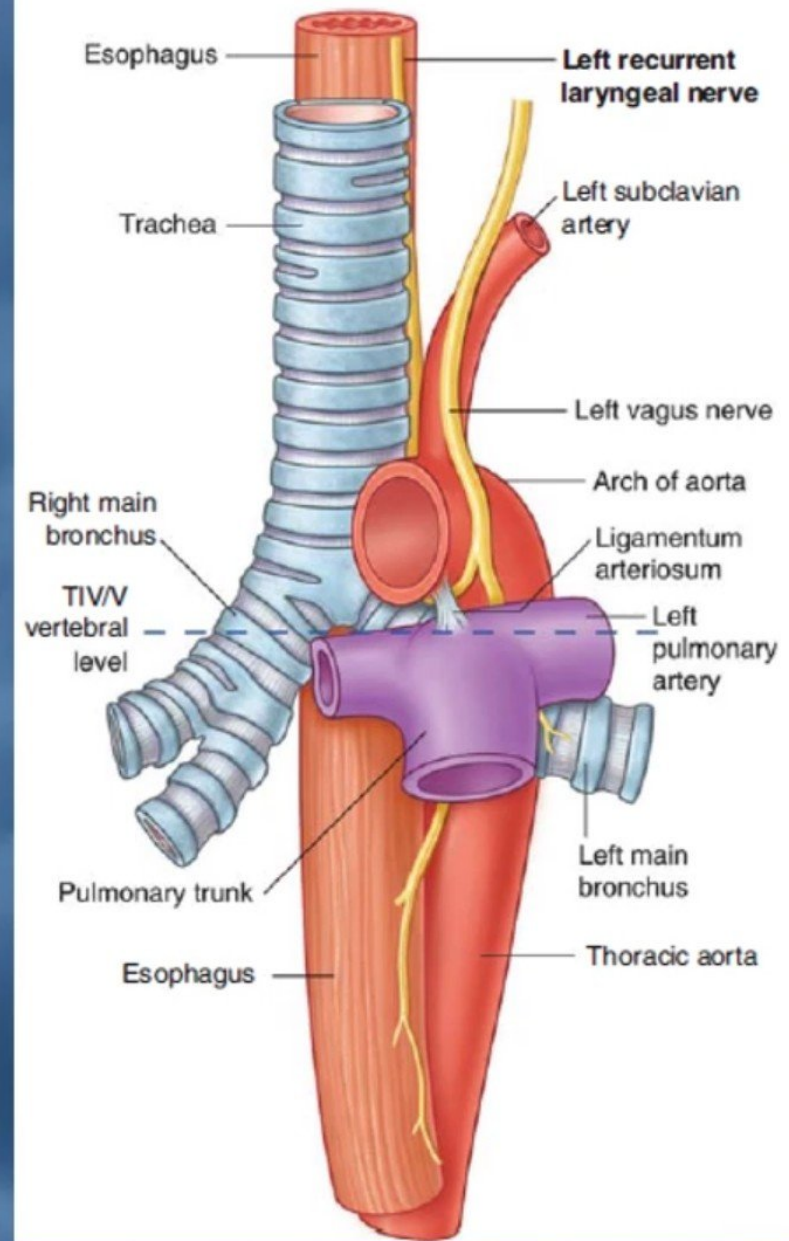
Introduction of Esophagus and course

In the thorax:

- It passes downward and to the left through **the superior** and **then the posterior mediastinum**.

At the level of the sternal angle:

- **the aortic arch pushes the esophagus over to the midline.**

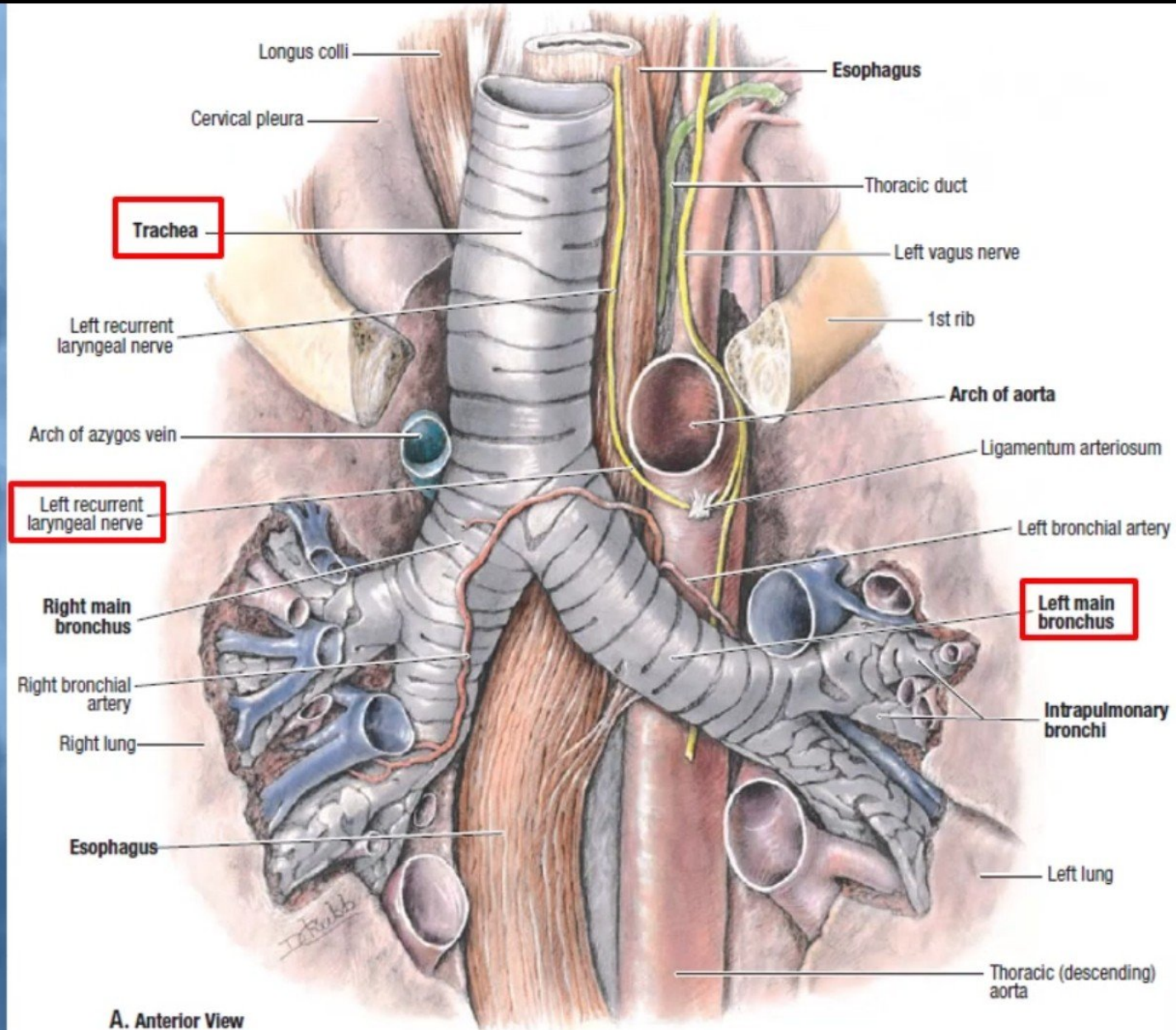


Relation of esophagus

The relations of the thoracic part of the esophagus from above downward are as follows:

Anteriorly:

- ✓ The trachea.
- ✓ The left recurrent laryngeal nerve.
- ✓ The left principal bronchus, (constricts it).
- ✓ The **pericardium**, which separates the esophagus from the **left atrium**.

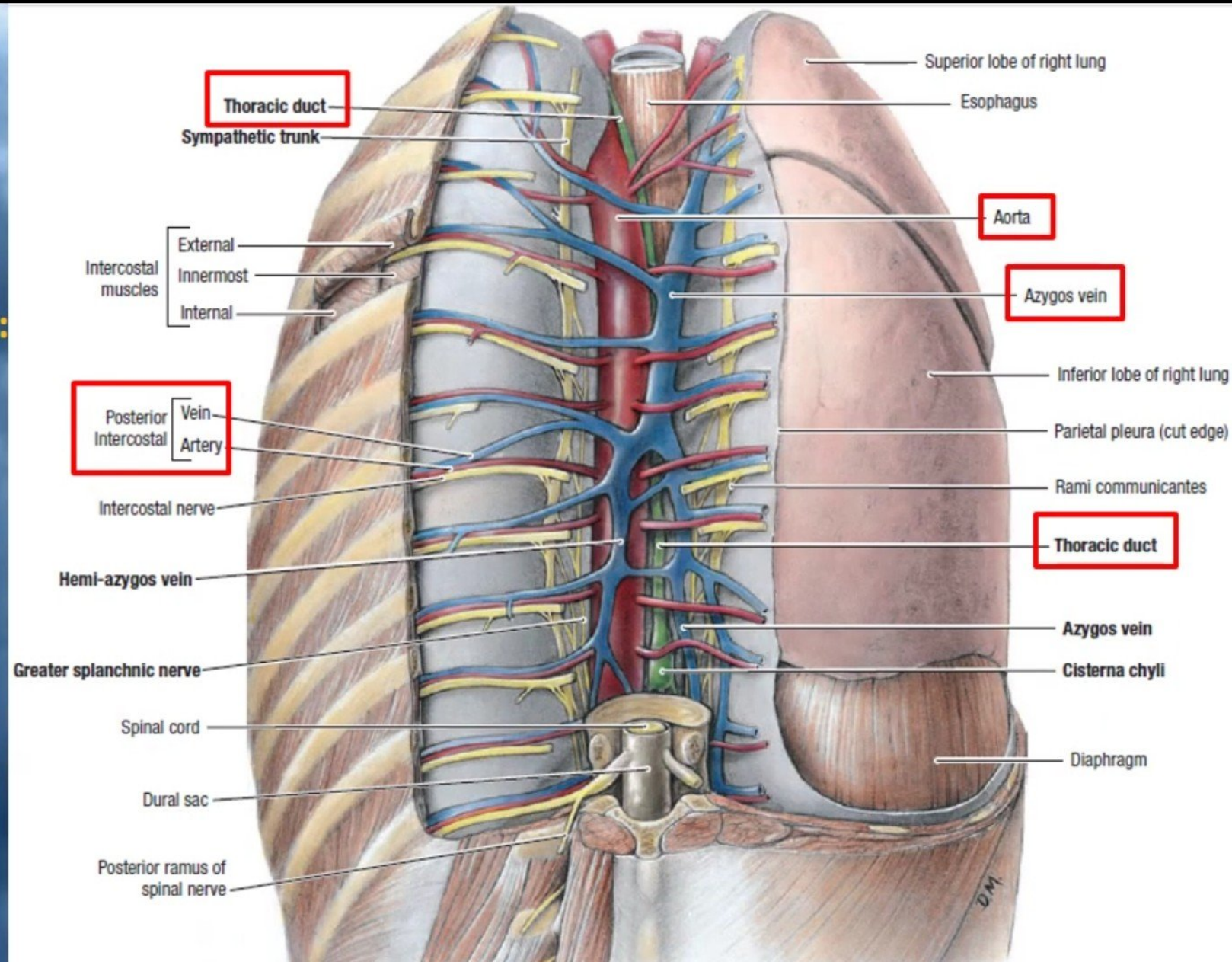


Relation of esophagus

The relations of the thoracic part of the esophagus from above downward are as follows:

Posteriorly:

- ✓ The **bodies of the thoracic vertebrae.**
- ✓ The **thoracic duct.**
- ✓ The **azygos veins.**
- ✓ The **right posterior intercostal arteries.**
- ✓ at its lower end, the **descending thoracic aorta.**

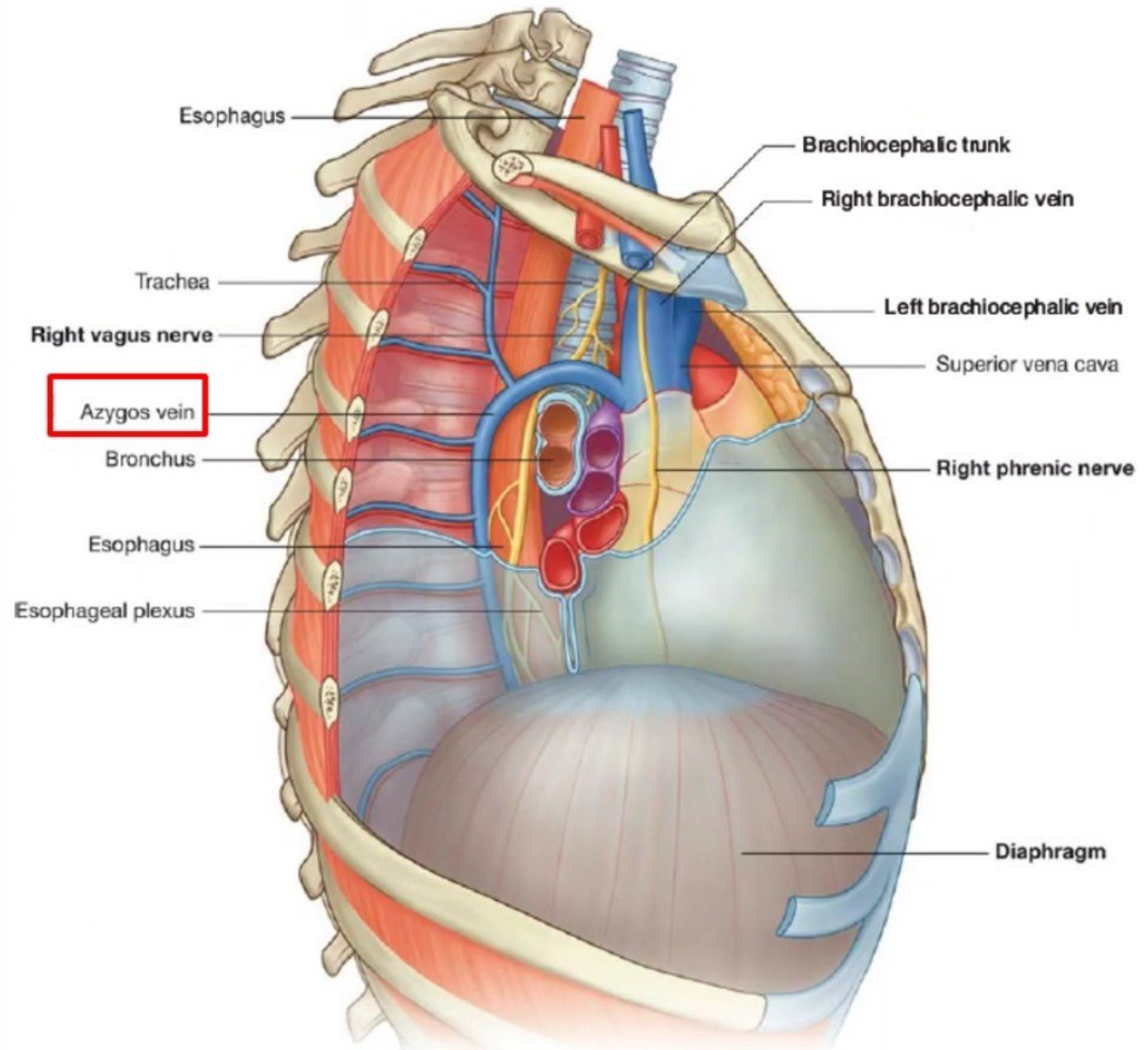


Relation of esophagus

The relations of the thoracic part of the esophagus from above downward are as follows:

Right side:

- The **mediastinal pleura**.
- The terminal part of the **azygos vein**.

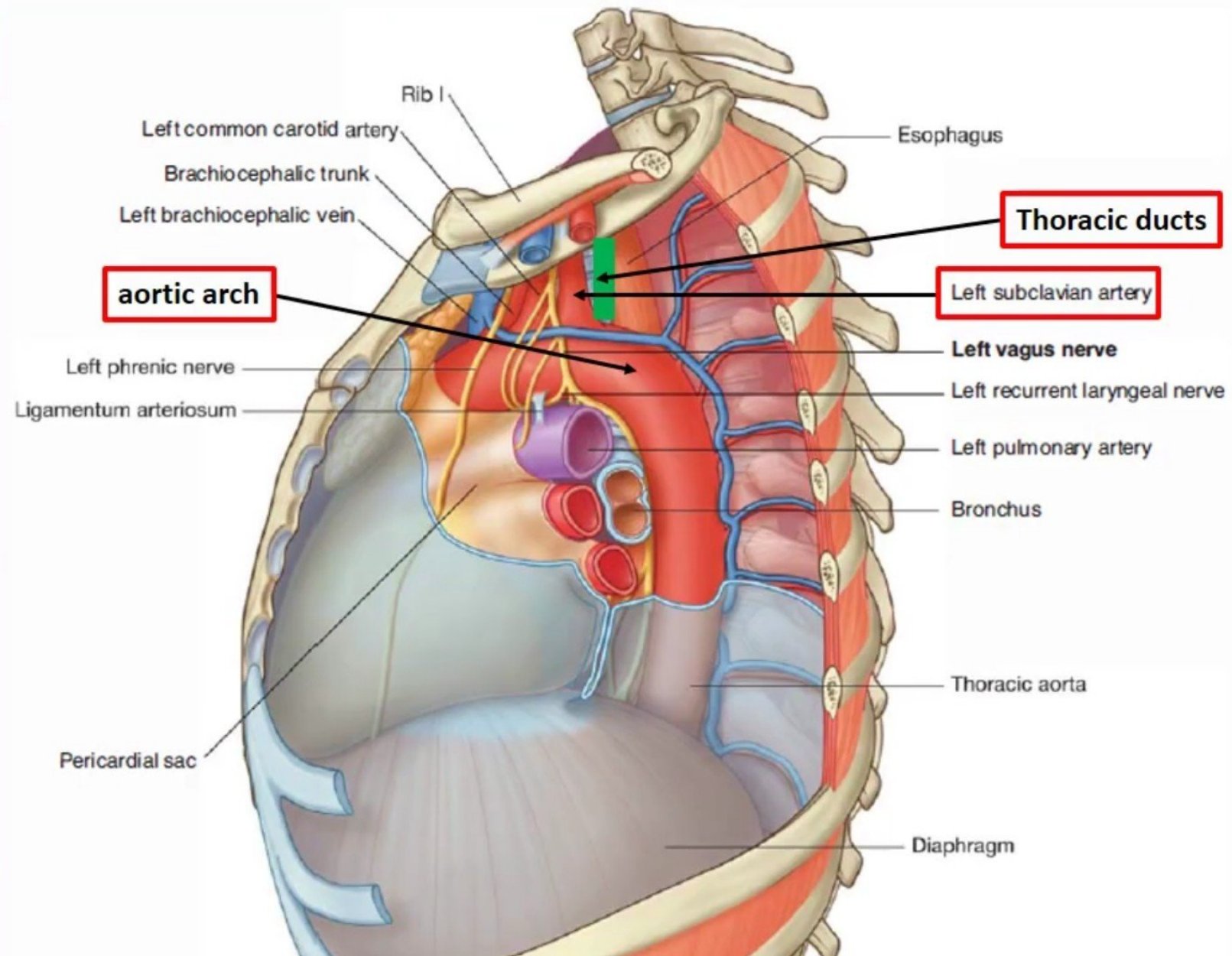


Relation of esophagus

The relations of the thoracic part of the esophagus from above downward are as follows:

Left side:

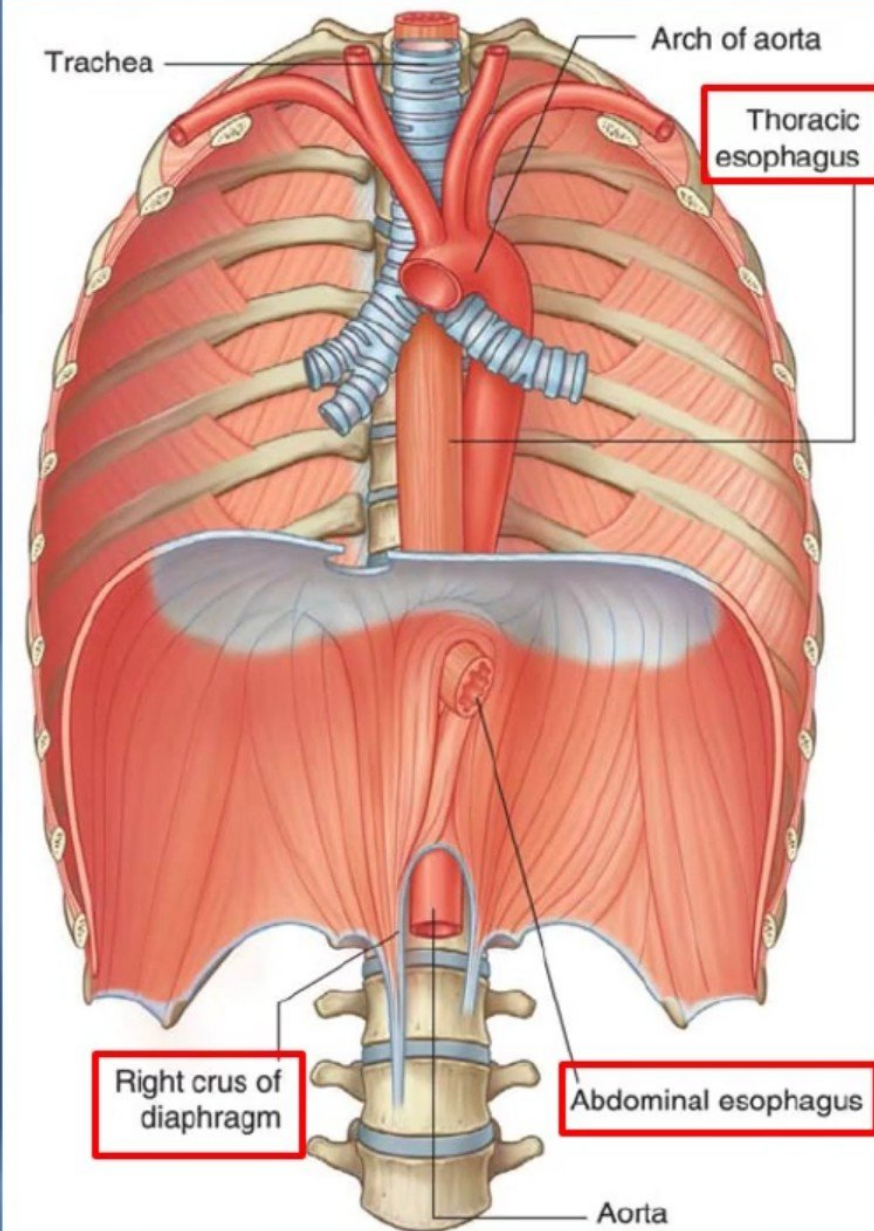
- The **left subclavian artery**.
- The **aortic arch**.
- The **thoracic duct**.
- The **mediastinal pleura**.



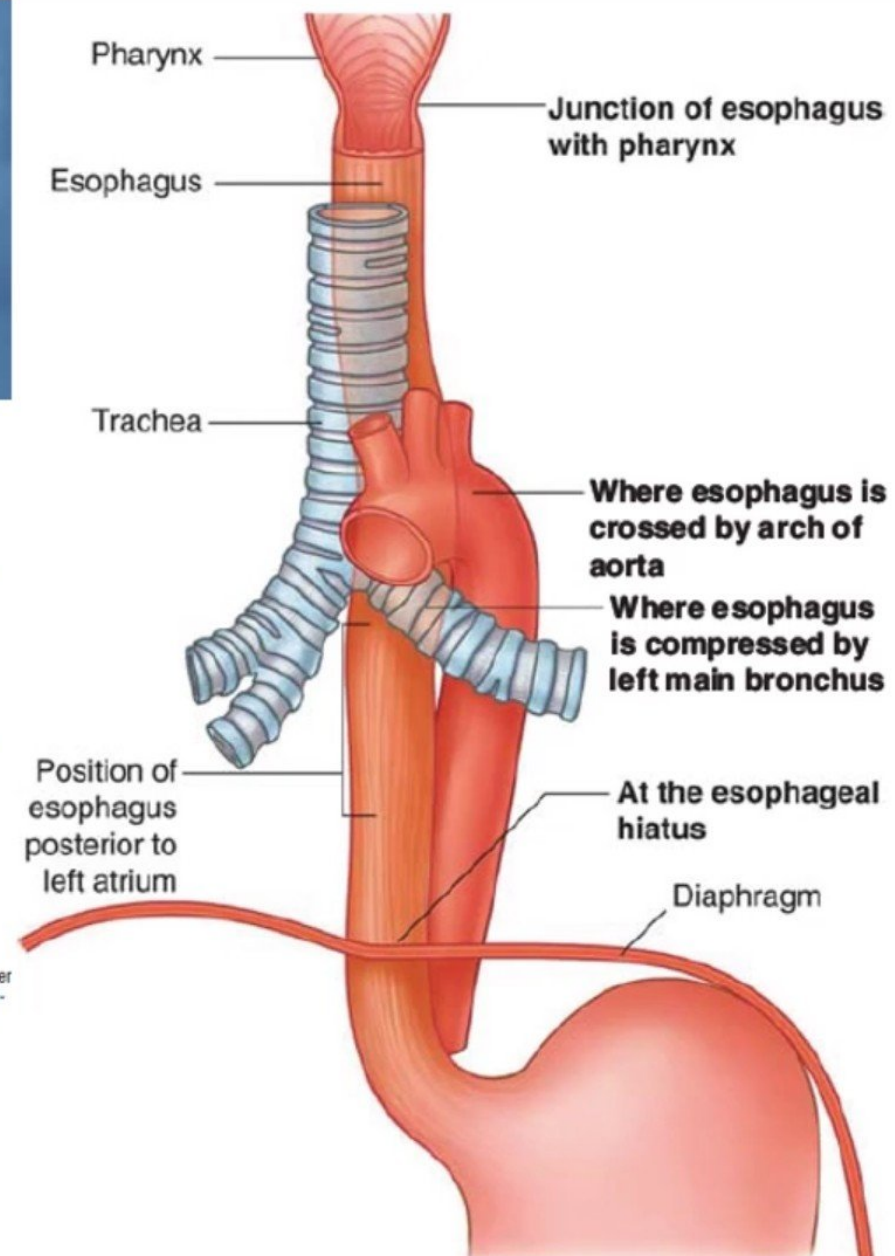
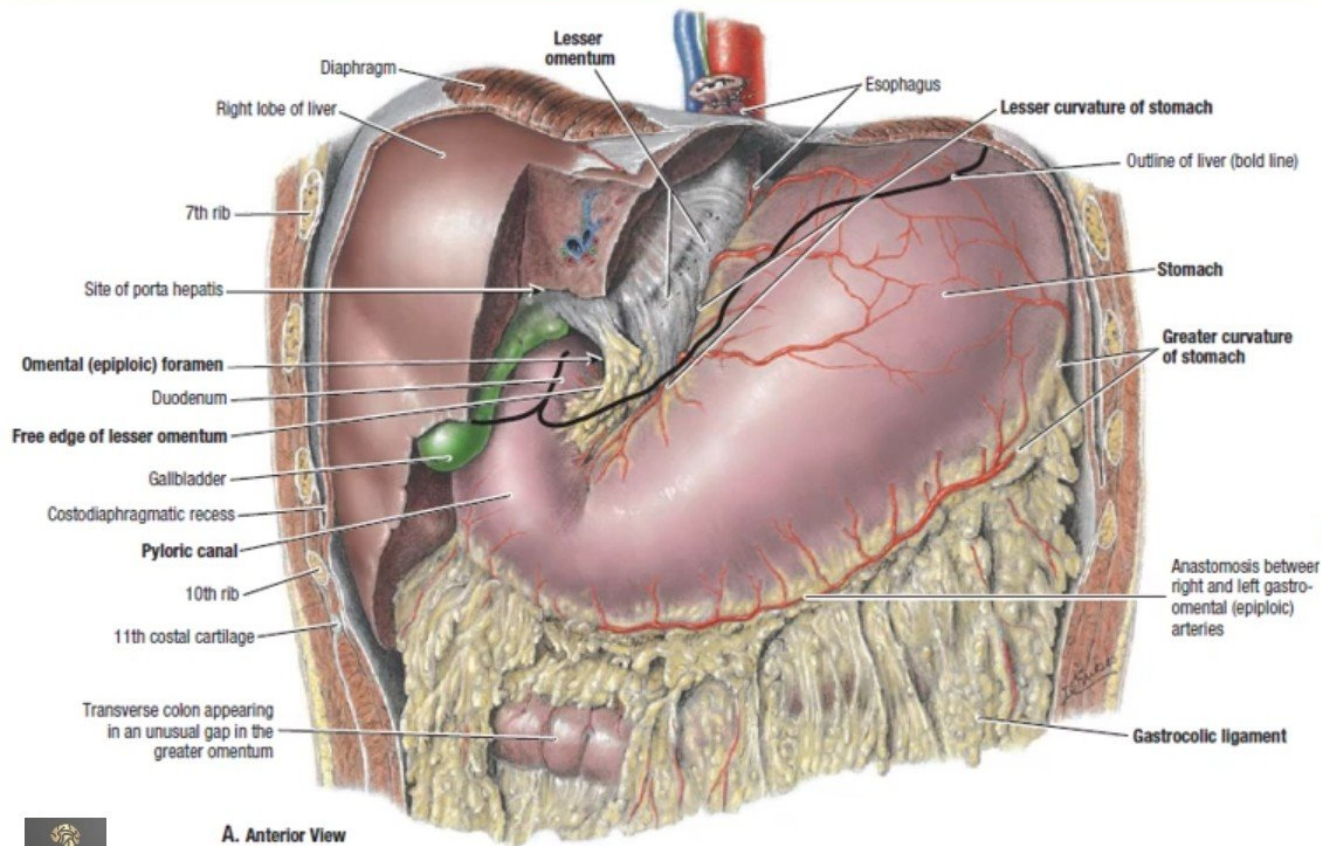
Abdominal Esophagus

The abdominal part of the esophagus:

- It extends from the **diaphragm** to the **cardiac orifice of the stomach**, entering the abdomen through an opening in **the right crus of the diaphragm**.
- It descends for about **0.5 in. (1.3 cm)** and then **enters the stomach**.
- It is related to **the left lobe of the liver anteriorly** and to the **left crus of the diaphragm posteriorly**.
- The **short abdominal part (½ in. long)**.
- at the **gastroesophageal junction**, the diaphragmatic musculature forming the esophageal hiatus functions as a **physiologic esophageal sphincter**.



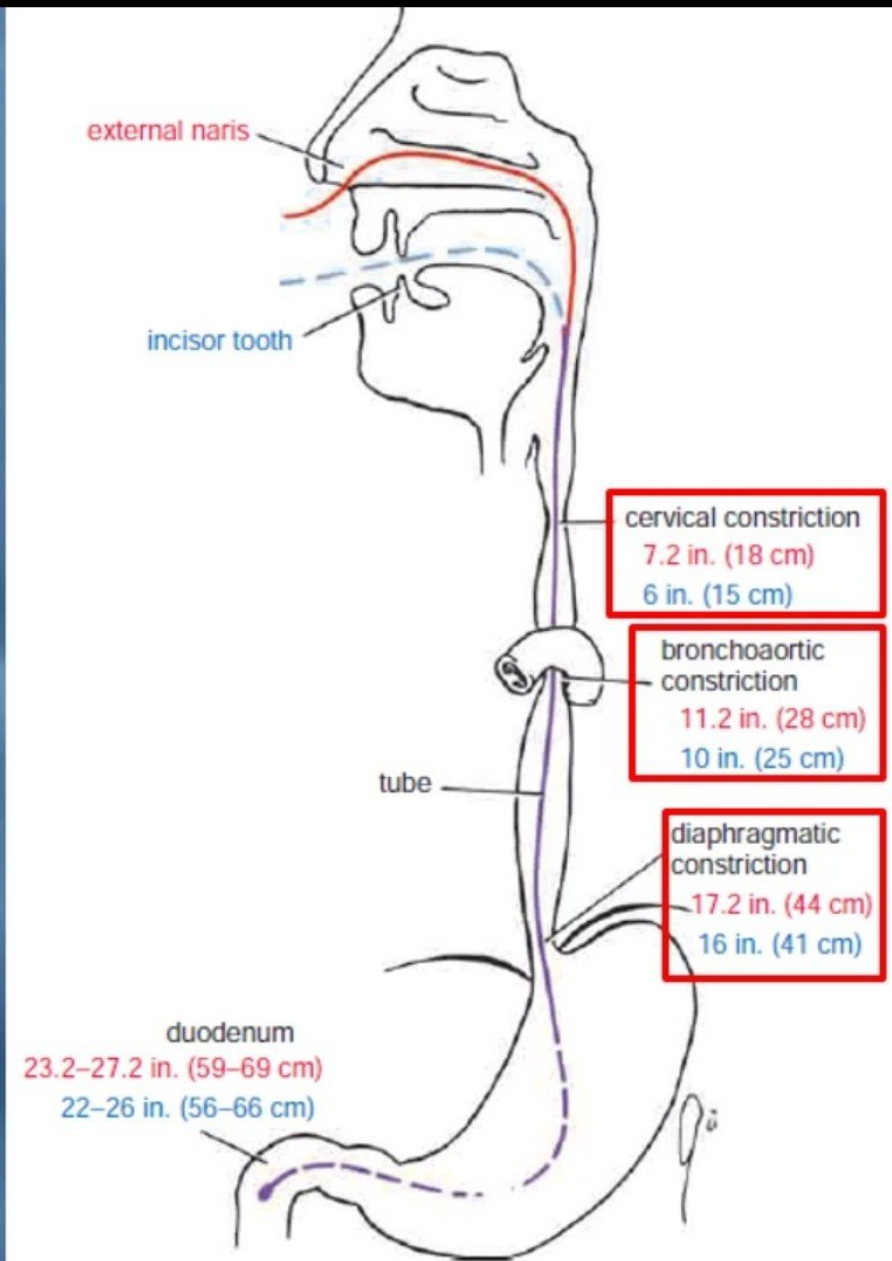
Esophagus



Esophageal constrictions

Has three constrictions:

- 1) Upper or pharyngoesophageal constriction at the **beginning of the esophagus** at the level of the **cricoid cartilage (C6)** is caused by the **cricopharyngeus muscle** (the **superior esophageal sphincter**).
 - 2) Middle or thoracic constriction by the **aortic arch** and then **left main bronchus**.
 - 3) Inferior or diaphragmatic constriction at the **esophageal hiatus** of the diaphragm (**T10**).
- The **left atrium** also presses against **the anterior surface of the esophagus**.



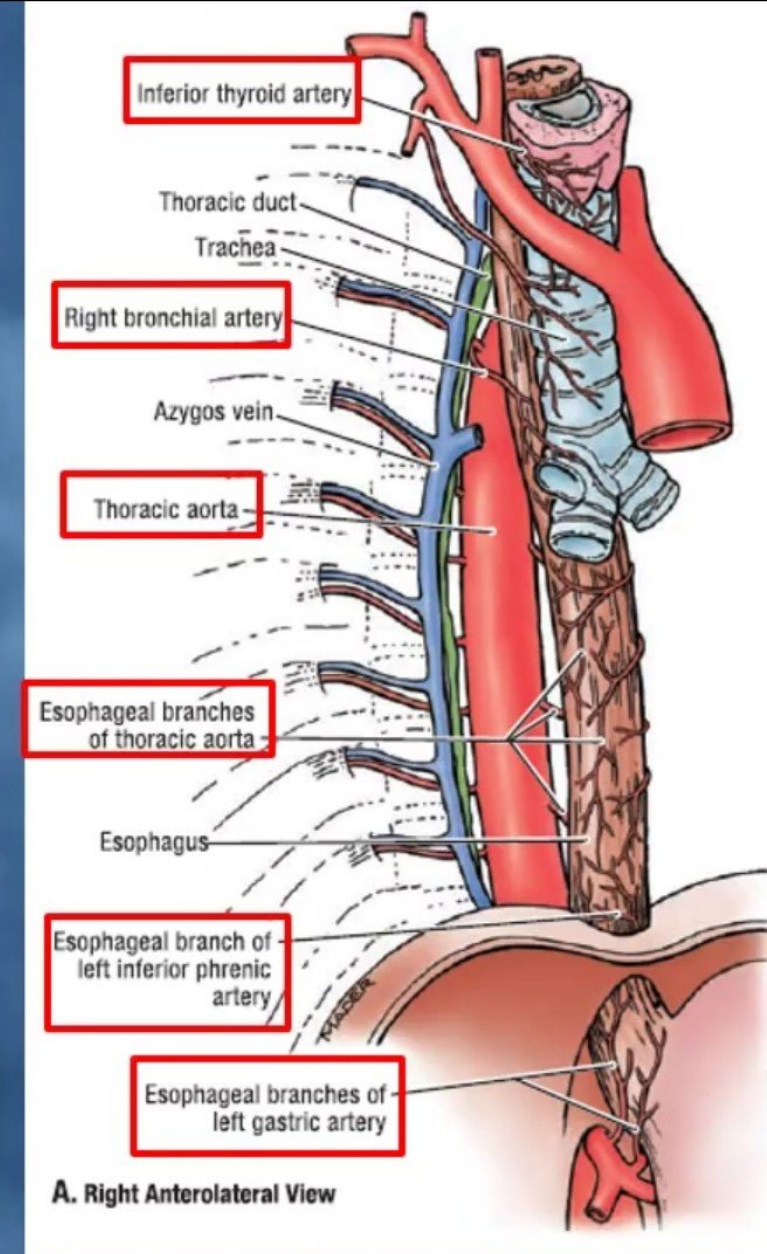
Blood Supply of Esophagus

The arterial supply of the esophagus:

- 1) The upper third of the esophagus is supplied by **the inferior thyroid artery** in the neck.
- 2) The middle third by branches from the branches of the **descending thoracic aorta** (bronchial, and esophageal arteries).
- 3) The lower third by branches from **the left gastric artery** and **the left inferior phrenic artery**.

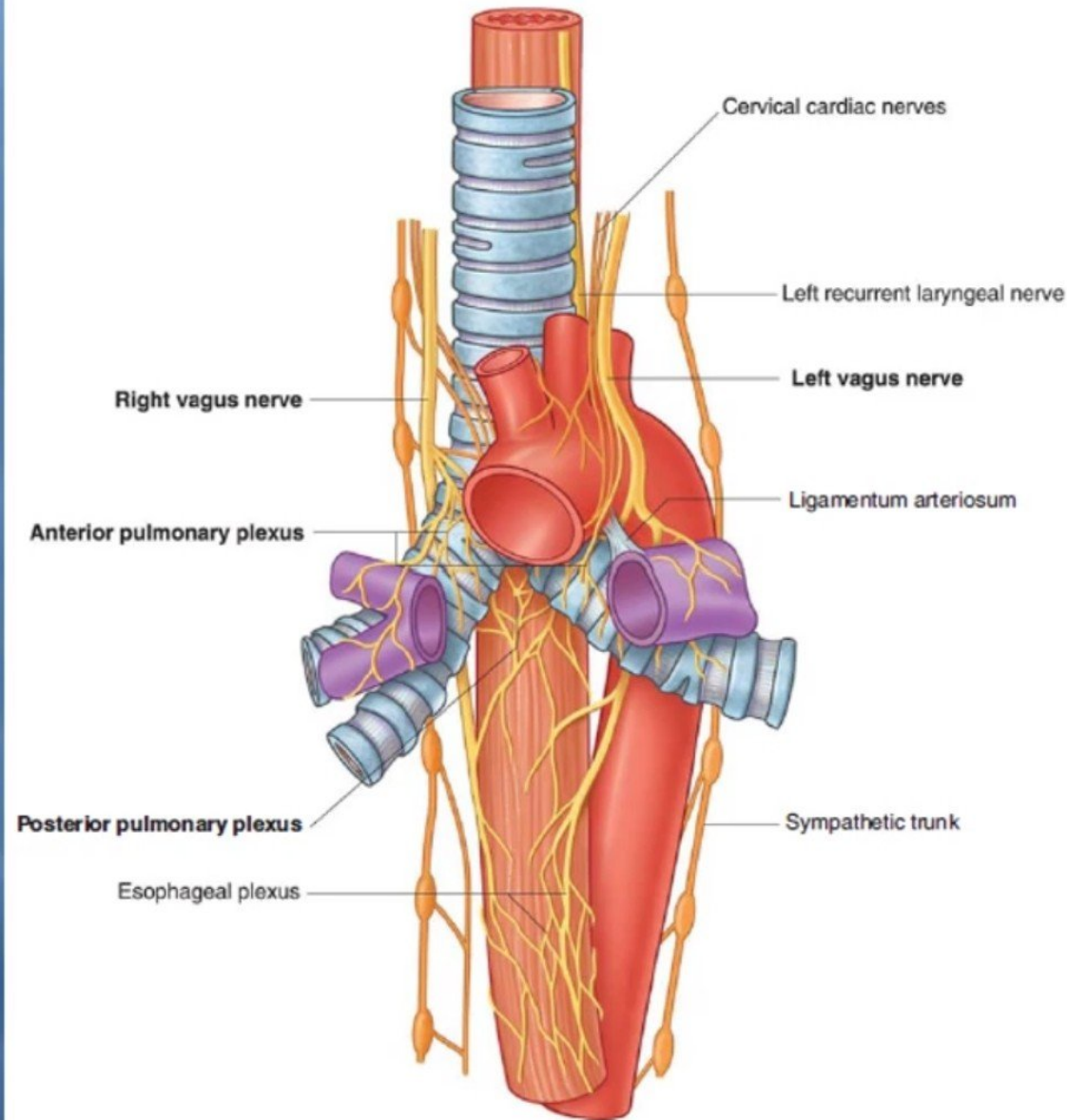
The venous drainage of the esophagus:

- The veins from the upper third drain into **the inferior thyroid veins**.
- The veins from the middle third drain into **the azygos veins**.
- The veins from the lower third into **the left gastric vein**, (tributary of the portal vein).



Innervation of esophagus

- The esophagus is supplied by **parasympathetic** and **sympathetic**.
- **efferent** and **afferent** fibers via **the vagi** and **sympathetic trunks**.
- In the lower part of its thoracic course, the esophagus is surrounded by the **esophageal nerve plexus**.



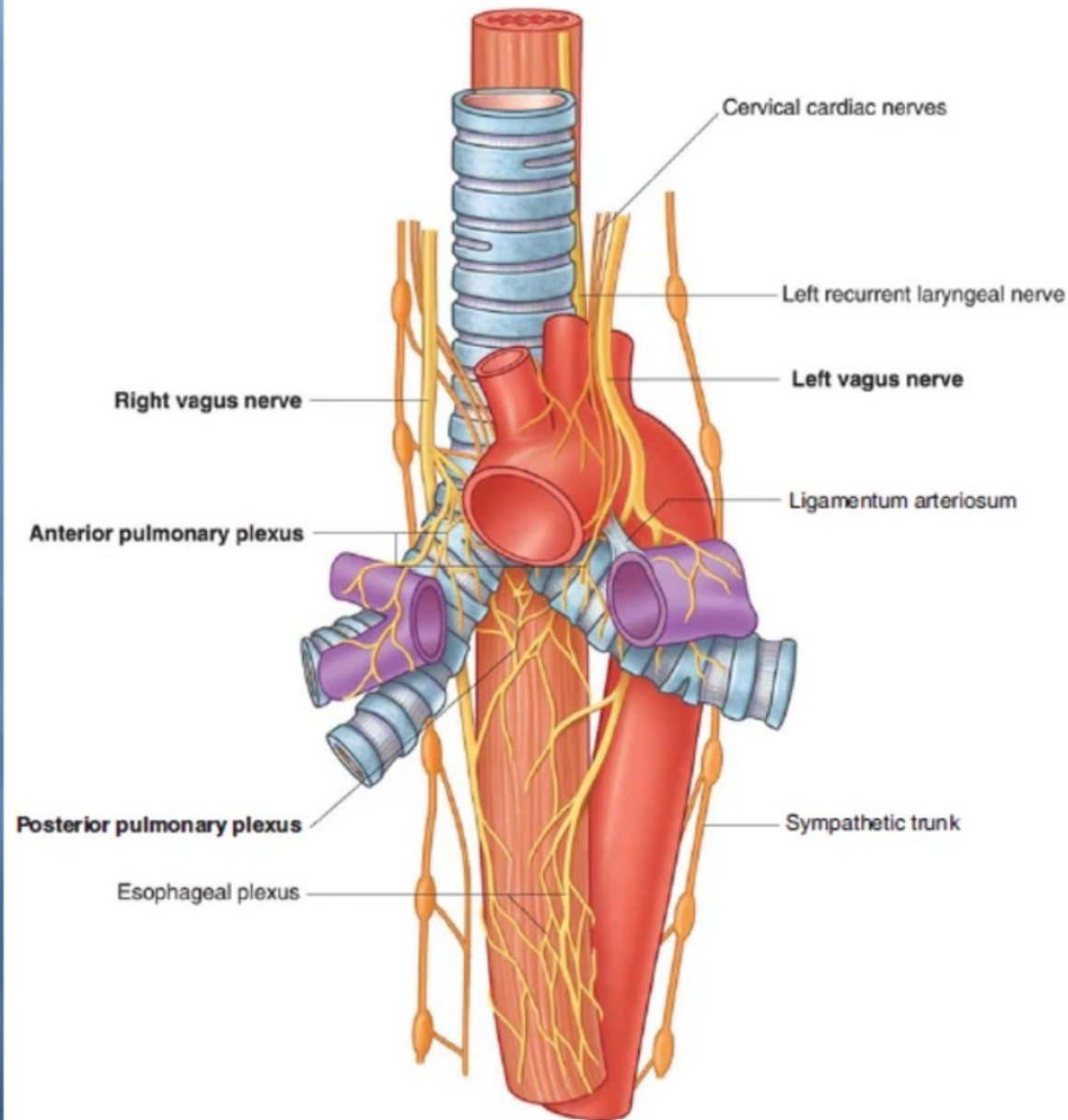
Innervation of esophagus

Inferiorly to the level of the roots of the lungs,

- The **vagus nerves leave** the **pulmonary plexus** and **join** with **sympathetic nerves** to form the **esophageal plexus**.
- The **left vagus** lies **anterior to the esophagus**.
- The **right vagus** lies **posterior to the esophagus**.

At the opening in the diaphragm, the esophagus is accompanied by:

- The **two vagi**.
 - The branches of the **left gastric blood vessels**, and
 - The **lymphatic vessels**.
- ✓ Fibers from **the right crus of the diaphragm** pass around the esophagus in the form of a sling.



Lymphatic drainage of esophagus

- Lymph vessels from the upper third of the esophagus drain into the **deep cervical nodes**.
- Lymph vessels from the middle third into **the superior and posterior mediastinal nodes**.
- Lymph vessels from the lower third into nodes along **the left gastric blood vessels** and the celiac nodes.

Function of esophagus and Gastroesophageal Sphincter

Function

- The esophagus **conducts food** from **the pharynx** into **the stomach**.
- **Wavelike contractions** of the muscular coat, called **peristalsis**, propel the food onward.

Gastroesophageal Sphincter

- **No anatomic sphincter** exists **at the lower end of the esophagus**. However, **the circular layer of smooth muscle** in this region serves as a **physiologic sphincter**.
- As the food descends through the esophagus, **relaxation of the muscle at the lower end** occurs ahead of **the peristaltic wave** so that the food enters the stomach.
- The **tonic contraction** of this sphincter **prevents the stomach contents** from **regurgitating into the esophagus**.
- The **closure** of the sphincter is **under vagal control**, and this can **be augmented by the hormone gastrin** and reduced in response to **secretin, cholecystikin, and glucagon**.

Will be continue...